Thakur Shree DPS College of Engineering & Management, Vasai (East)

First Year Engineering

Academic Year: 2026-2027

Assignment No: 02

Subject: BSC102/AP **Date: 11/09/2026**

CO2: To explain the basic working principle of Optical fiber and its use in communication technology

Q. No.	Questions	Marks
1	A step-index glass fiber has a core refractive index $n_1 = 1.54$ and a cladding refractive index $n_2 = 1.48$. (a) Calculate the critical angle (θ_c) at the core–cladding interface. (b) If a portion of the cladding is stripped away and the unclad core is exposed directly to ambient water ($n_{\text{water}} = 1.333$), calculate the new critical angle (θ'_c) at the core–water interface.	**2**
2	An optical fiber is manufactured with a core refractive index of $n_1 = 1.475$ and is designed to have a fractional refractive index change of $\Delta = 1.35\%$. (a) Determine the exact refractive index of the cladding (n_2). (b) Determine the Numerical Aperture (NA) using both the exact formula and the approximate formula $n_1\sqrt{2\Delta}$, and compute the percentage error between the two values.	**3**
3	A step-index fiber has a core index $n_1 = 1.50$ and a cladding index $n_2 = 1.45$. (a) Find the maximum acceptance angle (θ_0) when light is coupled into the fiber core from air. (b) Calculate the new acceptance angle (θ'_0) if the fiber entrance terminal is submerged in ethyl alcohol ($n'_0 = 1.36$).	**2**
4	An optical signal of power $P_{\text{in}} = 2.5$ mW is coupled into a silica fiber link of length $L = 12$ km. The output power detected at the far end of the link is measured to be $P_{\text{out}} = 125\ \mu$W. (a) Calculate the overall attenuation (loss) in decibels (dB) across the link. (b) Determine the attenuation coefficient (α) of the fiber in dB/km. (c) If the minimum detectable power at the receiver is $P_{\text{min}} = 10\ \mu$W, determine the maximum transmission distance (L_{max}) possible before requiring a repeater.	**3**

NB: Science is a way of trying not to fool yourself. The first principle is that you must not fool yourself, and you are the easiest person to fool.

- Richard Feynman